

Sheet1

Minimal Energy found from:			
Thomson Problem Applet		CirclePack	
N=	Energy	CP energy	
5	6.474691494688	6.474691494688	0.000000000000
7	14.452977414220	14.452977414221	0.000000000000
10	32.716949460150	32.917158621973	-0.006119432439
19	135.089467556700	135.707490999340	-0.004574919524
20	150.881568333800	152.233263608944	-0.008958650749
40	660.675278834600	661.034060486276	-0.000543052939
42	732.078107543700	732.285573683192	-0.000283393449
50	1055.182314726000	1056.651311247700	-0.001392173183
60	1543.813496757000	1545.985871750810	-0.001407148596
80	2805.355875981000	2811.431137268190	-0.002165593798
93	3829.844338421000	3835.744744308890	-0.001540638566
100	4447.695739176000	4453.923855682790	-0.001400301835
117	6146.342446579000	6158.045004876350	-0.001903987355
148	9958.406004270000	9973.252233038770	-0.001490823809
150	10236.255527250000	10255.533096950100	-0.001883263821
176	14199.354775630000	14222.377158354300	-0.001621368230
200	18438.842717530000	18466.455031956100	-0.001497507997
200	18438.842717530000	18467.531542741800	-0.001555890771
201	18627.591226470000	18660.041340589900	-0.001742045642
250	29067.877673910000	29101.078051760500	-0.001142167248
252	31567.440234700000	29582.750853789600	0.062871406936
296	40996.413908530000	41043.292011571500	-0.001143468381
300	42131.264137310000	42187.575108950200	-0.001336560219
331	51455.475450870000	52950.028093194400	-0.029045551114
333	52089.255151000000	53314.927147549200	-0.023530226973
350	57634.596395220000	57707.815333651100	-0.001270399083
362	61719.051909870000	61780.067678784100	-0.000988605091
400	75582.448512210000	75681.861478466700	-0.001315291688
441	92126.484273140000	92194.389594371300	-0.000737087948
443	92975.636464930000	93080.804262865200	-0.001131132864
450	95978.441376190000	96097.838792300400	-0.001244002449
463	101682.905435200000	101790.514414015000	-0.001058279938
500	118824.414964100000	118934.398724137000	-0.000925599003
525	131168.973310600000	131257.720104578000	-0.000676583736
667	212930.287286000000	213092.489723447000	-0.000761763108
723	250626.957152500000	250762.909147511000	-0.000542447614
842	340985.657638700000	341252.910549389000	-0.000783765841
943	428628.162420200000	428868.225034711000	-0.000560071959
1086	569926.306413100000	570169.136551896000	-0.000426072873
2000	1950575.638939000000	1951262.883334200000	-0.000352329016
2768	3837358.209172000000	3796639.881337670000	0.010611031239 +
3000	4409208.494179000000	4410257.626678290000	-0.000237941232
4000	7860245.950333000000	7862495.386465180000	-0.000286178848
5000	12304605.643090000000	12310053.486388400000	-0.000442748306
6000	17743132.833970000000	17749578.244670900000	-0.000363262269
7000	24176599.047940000000	24182932.631965000000	-0.000261971670
8000	31980124.516550000000	31609491.940675400000	0.011589466316 +
9000	40028059.602020000000	40035503.958311600000	-0.000185978445
10000	49447282.426660000000	49470032.177823500000	-0.000460080919
20000	198437048.316900000000	198454524.357937000000	-0.000088068439

Energy for random point sets found from:

N=	Thomson Problem Applet Energy	CirclePack CP energy	(E – CP)/E
10	37.374771178700	33.145201092859	0.113166447645
20	180.337760416000	155.939573288205	0.135291616528
50	1275.896003445000	1156.390639445780	0.093663875172
100	4448.424872203000	4763.588015193500	-0.070848255741
148	10784.907034660000	10399.011578619300	0.035781064668
150	11123.474094700000	10673.575727042100	0.040445850265
200	19866.050073520000	19155.573259975300	0.035763365687
250	31311.603392860000	30015.684015188800	0.041387831898
252	29543.522868140000	30597.868645453300	-0.035687882654
300	45063.101698590000	43158.485966921700	0.042265526781
333	55299.189557950000	53314.927147549200	0.035882305442
350	61186.614106230000	59033.085228676500	0.035196078571
400	79865.272731570000	77521.797719603100	0.029342853681
441	97003.062205980000	94256.896070572600	0.028310097361
450	100360.415631900000	98480.319281673500	0.018733445237
500	124915.498210100000	121532.051173337000	0.027085886741
525	136974.989818300000	133381.420266906000	0.026235224081
667	221353.815449900000	218086.956401281000	0.014758539590
723	262502.728981900000	256981.456736536000	0.021033199414
842	354911.192104900000	346921.619096323000	0.022511471000
943	444033.662669000000	436404.772184610000	0.017180883176
1086	591598.063383400000	580593.184971283000	0.018601951381
2000	2001098.502882000000	1975747.957184550000	0.012668314759
2768	3837358.209172000000	3796639.881337670000	0.010611031239
3000	4500613.181970000000	4449656.600216450000	0.011322142049
4000	8007565.199837000000	7928463.537649290000	0.009878366297
5000	12495687.402190000000	12403120.887521100000	0.007407876949
6000	17992647.361880000000	17861055.166490300000	0.007313665007
7000	24500834.813420000000	24342748.499041900000	0.006452282772
8000	31980124.516550000000	31801746.014283200000	0.005577792612
9000	40491697.968900000000	40269588.654152900000	0.005485305035
10000	50014962.113380000000	49740085.271734300000	0.005495892230
20000	199969144.871000000000	199237072.917897000000	0.003660924557